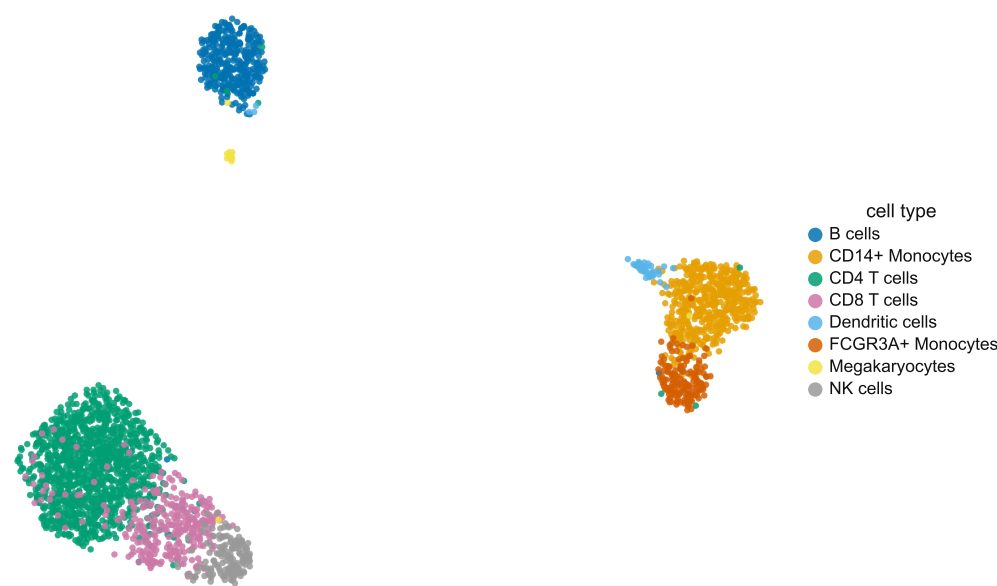


Figure 2 · The right ruler resolves cell types without a resolution knob · pbmc3k

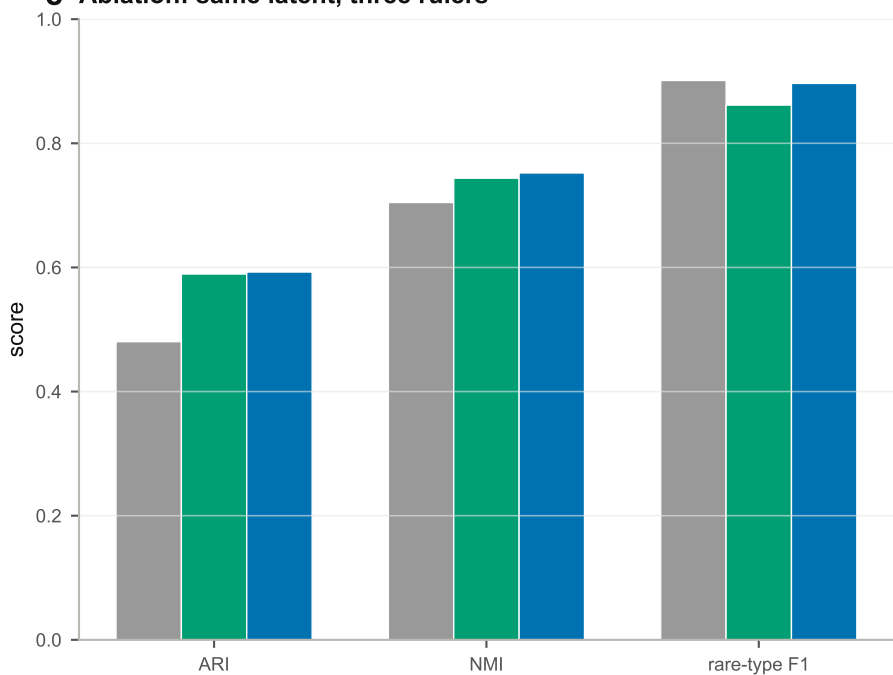
a Euclidean latent → UMAP



b Fisher-Rao graph → UMAP



c Ablation: same latent, three rulers



inter-cell ruler

- Euclidean  $L_2$
- $\text{pullback } J^T J$
- Fisher-Rao  $J^T \frac{1}{\text{Var}} J$

d Rare-type rescue (per cell type)

